



Introduction to
Internet of Things
Assignment-Week 6

TYPE OF QUESTION:MCQ/MSQ

Number of questions:15

Total marks: 15 X 1= 15

QUESTION 1:

State True or False.

Statement: “Python is popular for embedded application development as it is a very lightweight programming language.”

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: Python is popular for embedded application development as it is a very lightweight programming language.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 1:22)



QUESTION 2:

State True or False.

Adafruit provides a library to work with DHT22 Sensor.

a. True

b. False

Correct Answer: a. True

Detailed Solution: Adafruit provides a library to work with DHT22 Sensor. (Please refer to lecture Implementation of IoT with Raspberry Pi- II @ 4:41)

QUESTION 3:

Consider the following piece of Python code. What is the output?

```
x = [4, 5, 6]
y = [str(x[0] + 1), str(len(x) * 2) + '&Code']
z = y[1].split('&')
print(z[1])
```

a) 5

b) 12

c) Code

d) &Code

Correct Answer: c. Code

Detailed Solution:

1. `x = [4, 5, 6]`: A list with three elements.
2. `y = [str(x[0] + 1), str(len(x) * 2) + '&Code']`:



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- o $x[0]$ is 4, so $x[0] + 1$ is 5, and $\text{str}(x[0] + 1)$ becomes "5".
 - o $\text{len}(x)$ is 3, so $\text{len}(x) * 2$ is 6, and $\text{str}(\text{len}(x) * 2) + '&\text{Code}'$ becomes "6&Code".
 - o Therefore, $y = ["5", "6&\text{Code}"]$.
3. $z = y[1].\text{split}('&')$:
- o $y[1]$ is "6&Code".
 - o Splitting "6&Code" by '&' gives ['6', 'Code'].
4. $z[1]$ is 'Code'.
- Thus, the $\text{print}(z[1])$ statement outputs Code.

QUESTION 4:

State True or False.

Statement: "To indicate different blocks of code, Python follows rigid indentation."

- a. True
- b. False

Correct Answer: a. True

Detailed Solution: To indicate different blocks of code, Python follows rigid indentation.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 7::29).

QUESTION 5:

What is the output of the following line of code in Python?

```
>>> print "Hi, Welcome to python!"
```



- a. Hi, Welcome to python!
- b. “Hi, Welcome to python!”
- c. Hi, Welcome to python
- d. None of these

Correct Answer: a. Hi, Welcome to python!

Detailed Solution: The output of the following line of code in Python -

```
>>> print “Hi, Welcome to python!”
```

Output: Hi, Welcome to python!

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 07:31)

QUESTION 6:

During remote server access by a Raspberry Pi, where the Raspberry Pi acts as a client, the client needs the following?

- a. Only IP address of server
- b. Only port number
- c. Both server IP address and port number
- d. Client’s IP address

Correct Answer: c. Both server IP address and port number

Detailed Solution: A client can communicate with a server only if both IP address and port numbers are known. (Please refer Lecture 31@14:13)



QUESTION 7:

State whether the following command to install the PIL library is correct or not.

sudo pip install pillow

- a. Correct
- b. Incorrect

Correct Answer: a. Correct

Detailed Solution: The command to install the PIL library is *sudo pip install pillow*.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 17:40)

QUESTION 8:

What is the purpose of the "w" mode in the `open()` function in Python?

- A) To read a file
- B) To write data to a file, overwriting existing content
- C) To append data to a file
- D) To open a file in read and write mode

Correct Answer: B) To write data to a file, overwriting existing content

Detailed Solution: "w" mode is used to write data to a file, overwriting existing content

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @05:05).



QUESTION 9:

What will be the output of the given Python program when reading from the file?

with open("PythonProgram.txt", "w") as file:

file.write("Writing data")

with open("PythonProgram.txt", "r") as file:

f = file.read() print('Reading from the file\n') print(f)

- A) Writing data
- B) Reading from the file**
- Writing data**
- C) Error: File not found
- D) None of the above

Correct Answer: B. Reading from the file

Writing data

Detailed Solution: Reading from the file

Writing data

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING-II @05:05).

QUESTION 10:

Can we configure Raspberry Pi as a File Server?

- a. Yes**
- b. No

Correct Answer: a. Yes



Detailed Solution: We can configure Raspberry Pi as a File Server.

See lecture INTRODUCTION TO RASPBERRY PI-I @ 02:46

QUESTION 11:

Which command is used to configure the Raspberry Pi for the camera module?

- A) `sudo camera-config`
- B) `sudo raspi-config`
- C) `sudo enable-camera`
- D) `sudo pi-setup`

Correct Answer: B) `sudo raspi-config`

Detailed Solution: `sudo raspi-config` is used to configure the Raspberry Pi for the camera module

See lecture INTRODUCTION TO RASPBERRY PI-II @ 18:44

QUESTION 12:

What is the final step after enabling the camera in the Raspberry Pi configuration?

- A) Restart the camera service
- B) Run a camera test command
- C) Reboot the Raspberry Pi
- D) Reinstall the Raspberry Pi OS

Correct Answer: C) Reboot the Raspberry Pi

Detailed Solution: after enabling the camera in the Raspberry Pi configuration, reboot.

See lecture IMPLEMENTATION OF IOT WITH RASPBERRY PI-II @ 18:44



QUESTION 13:

Which command Exits the nano editor?

- a. Ctrl + X
- b. Ctrl + O
- c. Ctrl + K
- d. None of these

Correct Answer: a. Ctrl + X

Detailed Solution: Ctrl + O exits the nano editor.

See lecture IMPLEMENTATION OF IOT WITH RASPBERRY PI-II @ 10:20

QUESTION 14:

In a temperature-controlled fan system using a relay, when should the fan turn on?

- A) When the relay is manually triggered
- B) When the surrounding temperature is lower than a predefined threshold
- C) When the surrounding temperature exceeds a predefined threshold
- D) When the battery voltage drops below a certain level

Correct Answer: C) When the surrounding temperature exceeds a predefined threshold

Detailed Solution: In a temperature-controlled fan system using a relay, the fan should turn on when the surrounding temperature exceeds a predefined threshold.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 11:18)



QUESTION 15:

What does the following line of code do?

raspistillcapture -o image.jpg

- a. Captures video feed
- b. Captures still image
- c. Both (a) and (b)
- d. None of these

Correct Answer: d. None of these

Detailed Solution: Command is wrong.

(Please refer to lecture INTRODUCTION TO RASPBERRY PI-II @ 19:29)

*******END*******